

Self-Attention

Definition

Self-attention transforms input word embeddings into contextual embeddings by allowing each token to attend to all other tokens in the sequence and determine which tokens are most relevant to its meaning.

The output embedding of a word is no longer based only on the word itself but also on the context provided by other words in the sentence.

What Does Self-Attention Determine?

Self-attention determines:

- Which words are important for understanding the current word.
- How much attention should be given to each word in the sequence.
- The contextual meaning of a word based on surrounding words.
- Long-range dependencies between words.

Example:

Sentence: "The animal didn't cross the street because it was tired."

For the word "**it**", self-attention assigns higher attention to "**animal**" than "**street**", helping the model understand that "it" refers to the animal.

Query, Key, and Value

For every input embedding X , three vectors are generated:

$$Q = XW_Q$$

$$K = XW_K$$

$$V = XW_V$$

where:

- Q = Query
 - K = Key
 - V = Value
 - W_Q, W_K, W_V are learnable weight matrices
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Attention Score

The similarity between words is calculated using the dot product of Query and Key:

$$Score = QK^T$$

A larger score indicates stronger relevance between two words.

Scaled Dot-Product Attention

To prevent large dot-product values from causing unstable softmax outputs, the scores are scaled:

$$\frac{QK^T}{\sqrt{d_k}}$$

where:

- d_k = dimension of the Key vector
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Attention Weights

Softmax converts the scaled scores into probabilities:

$$Attention\ Weights = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)$$

These weights indicate how much importance each word receives.

Final Self-Attention Formula

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

This formula computes a weighted sum of the Value vectors to generate contextual embeddings.

Output

The resulting output is a contextual embedding that captures:

- Word meaning
- Sentence context
- Relationships between words
- Long-range dependencies

One-Line Interview Answer

Self-attention computes attention scores between all tokens using Query, Key, and Value vectors, determines the importance of each token through attention weights, and produces contextual embeddings using:

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

allowing the model to understand words based on their surrounding context.